GAS PROCESSING & METHANOL COMPLEX PROJECT NO. 11998A

Project	FEED for Gas Gathering Pipelines & Associated Infrastructure
Client	Brass Fertilizer and Petrochemical Company Limited
Originating Company	Epic Atlantic Limited
Document Title	**Data Sheets for Mainline Valves**
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Data Sheets for Mainline Valves

R01	14.07.22	Issued For Review	C. Anthony	E.Nwachukwu	A. Duru	
Rev #	**Issue Date**	**Status Description**	**Originator**	**Reviewer**	**Approver (Epic)**	**Approver (BFPCL)**

Revision History

Rev No.	Date	Description of Revision
R01	14/07/2022	Issued for Review

Revision Philosophy:

a. All documents for review shall be issued at R01, with subsequent R02, R03, etc as required.

b. All documents approved for issue or approved for design shall be issued at A01 with subsequent A02, A03, etc. as required. (Management of Change is required for A02, A03, etc.).
for the subsequent project phases, post FEED, it is expected that:

c. All Detailed Design documents for review shall be issued at D01, with subsequent D02, D03, etc. as required.

d. All documents approved for construction shall be issued at C01 with subsequent C02, C03, etc. as required. (Management of Change is required for C02, C03, etc.)

e. All approved "As-Built" documents shall be issued at Z01, with subsequent Z02, Z03, etc as required. (Use versions Z01.1, Z01.2, Z01.3, etc to review the "As-Built" document to Z02).

TABLE OF CONTENTS

1. SCOPE

This document covers data sheets of Mainline (Pipeline) Valves within BBOU manifold Station, Elepa Manifold Station and Akabel Pigging Station. excluding piping valves.

This document shall be read in conjunction with the project reference documents listed bellow.

1.1 PROJECT REFERENCE DOCUMENTS

S/No	Document Number	Description
1	EA-BFPCL-GPMC-GGPL-FEED-MPP-SPC-003	Specifications for Mainline Valves
2	EA-BFPCL-GPMC-GGPL-FEED-PSE-DWG-003	Process Engineering Flow Sheets 1 - 8
3	EA-BFPCL-GPMC-GGPL-FEED-MPP-LST-010	Valve List.

1.2 CODES AND STANDARDS

S/No	Document Number	Description
1	API 6D	Pipeline Valves
2	API 607	Fire Test for Quarter-turn Valves and Valves Equipped with Nonmetallic Seats
3	ASME B16.5	Pipe Flanges and Flanged Fittings NPS ½ Through NPS 24
4	ASME B16.34	Valves – Flanged, Threaded, and Welding End
5	ASME B1.20.1	Pipe Threads, General Purpose (Inch)
6	BS EN ISO 6755-2	Testing of Valves - Part 2: Specification for Fire Type-Testing Requirements
7	EEMUA Publication No.182	Specification for integral block and bleed valve manifolds for direct connection to pipework
8	ISO 5208	Industrial valves - Pressure testing of metallic valves
9	ISO 14313	Petroleum and natural gas industries - Pipeline transportation systems - Pipeline Valves

2. ABBREVIATIONS

Abbreviation	Definition
AED	Anti-Explosive Decompression
API	American Petroleum Institute
ASME	American Society of Mechanical Engineers
ASTM	American Society for Testing and Materials
CA	Corrosion Allowance
CS	Carbon Steel
DPT	Dye Penetrant Test
EN	European Standards
ENP	Electro-less Nickel Plating
FB	Full Bore
GR	Grade
HT	Hardness Test
LCB	Low Temperature (Wrought) Carbon Grade B
LTCS	Low Temperature Carbon Steel
MPI	Magnetic Particle Inspection
MSS	Manufacturers Standardization Society
NDT	Non-Destructive Testing
NPT-F	National Pipe Taper - Female
OS&Y	Outside Screw & Yolk
P-T	Pressure-Temperature
PEEK	Polyetheretherketone
PS	Pigging Station
RB	Reduced Bore
RF	Raised Face
RTJ	Ring Type Joint
RPTFE	Reinforced Polytetrafluoroethylene
RT	Radiographic Testing
SPWD	Spiral Wound
SS	Stainless Steel
SW	Socket Welded
TGS	Terminal Gas Station
THRD	Threaded
UT	Ultrasonic Testing
WCB	Wrought Carbon Grade B

3. IMPORTANT NOTES

3.1 This data sheet shall be completely filled by the vendor, stamped, signed and deviations clearly marked for acceptance. Quotation without this requirement will not be considered for technical review and evaluation.

3.2 Gland packing design shall take into account the requirement for low emission, low maintenance service. Graphite packing shall be low to medium density, high purity incorporating a passivating corrosion inhibitor and anti-extrusion elements.

3.3 Valves shall have position indicator showing open and close positions.

3.4 Minimum thickness of weld overlay shall be 3.0mm undiluted thickness as per trim material in final machined condition.

3.5 The wall thickness of valve body and other pressure containing parts shall include corrosion allowance specified in the datasheet in addition to the minimum wall thickness required as per ASME B16.34

3.6 Hydrostatic test to be performed on body and seat, low pressure pneumatic seat test to be performed on seat.

3.7 Machined surfaces shall be subject to 100% MPI/DPT.

3.8 Valve stem shall be of sufficient length to accommodate a minimum of 100mm insulation thickness If Isulation is specified on the data sheet.Valve stem shall be anti-blow out stem design.

3.9 For 2" & above trunnion mounted ball valves, provision shall be made for drain. For 8" & above vent and drain connection shall be foreseen.Vent and drain connections shall be as per ASME B16.34 / MSS SP-45 and with double O' rings, unless specified otherwise.

3.10 Lifting eyes / lug shall be provided for all valves of sizes 8" NPS and larger valves weighing 250 Kg and above, whichever is stringent, to facilitate maintenance. Lifting lugs shall be independent of body bolting.

3.11 Vendor shall confirm suitability of seat and seal material based on service fluids and service design conditions. Vendor shall submit P-T rating chart of selected seats /seals complying with the above at the time of bid as well as post order. Requirement listed in the data sheet & specification shall be considered as minimum.

3.12 For the given temperature in the data sheet, valve seat insert (soft) material shall be able to withstand the limiting pressure indicated in rating table of ASME B16.34. Reduction in limiting pressure of valve due to limitation of seat insert material is not acceptable.

3.13 Valve bolting material shall be able to withstand the pressure as indicated in full rating pressure table of ASME B16.34. Vendor shall submit bolt strength calculation report for the same for review.

3.14 All soft seated SW ball valves shall supply with pup piece of 100 mm on both side of valve. Pup piece shall be welded by the valve manufacturer and under its responsibility prior to valve testing. 2 nos. of test rings shall be supplied by manufacturer per material and size.

3.15 All elastomeric materials (o-rings, seals, seats, etc.) shall be resistant to explosive decompression (ED) and resistant to absorption of hydrocarbon liquids.

3.16 Valve stems shall be one-piece design and made from bar / forged material.

3.17 Valves shall be tested and qualified for both vertical & horizontal positions as per applicable standards indicated in datasheet. Change of field installation position from the valve test position shall not be considered as a cause for valve leakage if any.

3.18 Stem & seat sealant injection connection shall be suitable for injecting sealant / grease with the valve in operating condition.

3.19 All sealant injection connections shall incorporate an internal non-return valve, a giant button head and a cover with vent holes that seals off the fitting by plugging the sealant port.

3.20 Valves shall not be shipped or tested with sealant injected into the valves.

3.21 For flanged-end valves, body and flanges shall be integrally forged.

3.22 Galvanised Bolts and Nuts used in valves shall be hot dip Per ASTM A153.

3.23 All threaded end valves used for instrument connections shall be NPT (Female) threaded to ASME B1.20.1.

3.24 All manual valves (whether operated with a lever or a gear operator) shall be provided with locking device plate to facilitate heavy duty padlocks or car-seals to prevent operation in the locked open and close position.

3.25 Color of Top Coat shall be Aluminium / Light Grey (RAL 9006 / 7035). Flange faces should not be painted. The Colour shall be discusses with Vendor before PO.

3.26 The use of asbestos material in any form is not permitted.

3.27 Valves shall have a double block and bleed feature, in the open and closed position to facilitate flushing, draining and venting of the valve.

3.28 Valve support shall be provided for valves weighing 250 kg or more; valve support material shall be same as that of valve body.

3.29 Vendor shall provide Inspection and Testing Plan based on the specifications and standards mentioned.

3.30 "O" ring groove dimensions and surface finish shall be in accordance with BS 4518. " O" rings shall be Anti Explosive Decompression certified for Gas service and compatible with methanol.

S/N	Datasheet 1: Datasheet for Ball Valve, Carbon Steel, SW with Pup Piece, CL 1500, 0.5" to 1.5", FB			
1	**Description**	**Specification Requirement**	**Vendor's Confirmation**	
2	**General Data**			
3	Piping Class	As Per Table 1.1 Below		
4	Valve Type	Ball Valve		
5	Valve Size Range	0.5" to 1.5"		
6	Valve ASME Class / Rating	1500#		
7	Valve Port	Full Bore		
8	Valve Service	Hydrocarbon Gas		
9	Valve End Connection	Socket Welded with 100mm Pup Piece (As Per Table 1.1 Below)		
10	Corrosion Allowance	As Per Table 2.1 Below		
11	Design Temperature (Min / Max)	- 29 / 150 ^{o}C		
12	Design Pressure	Full rating as per ASME B16.34		
13	**Design**			
14	Valve Operation	Hand Lever		
15	Body Construction	Split Body, Long Pattern		
16	Ball Support	Floating Ball		
17	Lifting Lugs	Not required		
18	Sealant Injection Facility	Not required		
19	Vent / Drain Requirements	Not required		
20	Locking Facility	Required		
21	Stem	Anti-Blow-Out Type		
22	Seat	Self-Relieving		
23	Cavity Pressure Relief	Self-Relieving Seat		
24	Anti-Static Design	Required		
25	**Materials**			
26	Body	ASTM A105N		
27	Cover / Bonnet	ASTM A105N		
28	Ball (Solid) / Trunnion	ASTM A182 316L		
29	Stem / Trim	ASTM A182 316L		
30	Pup Piece Material / Length	ASTM A106 Gr. B / 100mm (Note 5.14)		
31	Seat Insert	PEEK (Note 5.11)		
32	Primary / Secondary Seals	Elastomer O-rings, AED (Note 5.30) (Note 5.11)		
33	Gland Packing	Filled Graphite (Note 5.2) (Note 5.11)		
34	Anti-Static Device / Spring	INCONEL 718		
35 36	Gaskets	SPWD; SS304 Winding Graphite Filled; SS304 Inner Ring; CS Outer Ring		
37 38	Bolting	ASTM A193 Gr. B7; A194 Gr. 2H (Hot Dip Zinc Galvanized) ***(Note 5.22)***		
39	Hand Lever	Malleable Steel / Nodular Iron		
40	Test Ring	Required (Note 5.14)		
41	**Codes and Standards / Testing / Certification**			
42	Valve Design Code	As Per API 6D/DEP 31.36.00.30-Gen		
43	End to End Dimensions	As Per API 6D		
44	Fire Safe	As Per API 607		
45	Hydrostatic & Pneumatic Test	As Per API 598		
46	Low Pressure Gas Seat Test	As Per API 598 at 6 barg		
47	Leakage Test	Leak Rate "A" As Per ISO 5208		
48 49	NDT (MPI / DP / Radiography / UT)	As Per Section 9 of Specification for Mainline Valves (Doc No.: EA-BFPCL-GPMC-GGPL-FEED-PPL-DSH-003)		
50 51	Material Certification	As Per Section DEP 31.36.00.30-Gen. section 15.0		
52 53 54	Painting / External Coating	As Per Coating / Painting Materials Specification - Linepipes / Pipelines / Field Joints, Structures, and Equipment (Doc. No.: EA-BFPCL-GPMC-GGPL-FEED-MCM-SPC-010)		
55 56	Marking	As Per Section 10 of Specification for Mainline Valves (Doc No.: EA-BFPCL-GPMC-GGPL-FEED-PPL-DSH-003)		
57 58	Applicable Project Spec.	Specification for Mainline Valves (Doc No.: EA-BFPCL-GPMC-GGPL-FEED-PPL-DSH-003)		

59	Table 1.1 (As Per Valve List)									
60 61	S/N	Location	Pipe Class	Corrossion Allowance	Size	Rating	End Connection	Valve Tag	Line Number	Reference P&ID Number
62	1	EPEPA Manifold Station		6.0mm	3/4''	1500#	Socket Welded with 100mm Pup Piece	MV1056	ELEP-24''-P-012-X70-CL900	EA-BFPCL-GPMC-GGPL-FEED-PSE-DWG-003_05
63	2	Akabel Pigging Station		6.0mm	3/4''	1500#	Socket Welded with 100mm Pup Piece	MV1056	AKABP-36''-P-012-X70-CL900	EA-BFPCL-GPMC-GGPL-FEED-PSE-DWG-003_05

S/N	Datasheet 2: Datasheet for Ball Valve, Carbon Steel, Flanged RTJ, CL 900, 2", RB			
1	**Description**	**Specification Requirement**	**Vendor's Confirmation**	
2	**General Data**			
3	Piping Class	As Per Table 2.1 Below		
4	Valve Type	Ball Valve		
5	Valve Size Range	2"		
6	Valve ASME Class / Rating	900#		
7	Valve Port	Reduced Bore		
8	Valve Service	Hydrocarbon Gas / Open / Close Drain		
9	Valve End Connection	Flanged RTJ		
10	Corrosion Allowance	As Per Table 2.1 Below		
11	Design Temperature (Min / Max)	- 29 / 150 ^oC		
12	Design Pressure	Full rating as per ASME B16.34		
13	**Design**			
14	Valve Operation	Hand Lever		
15	Body Construction	Split Body, Long Pattern with Double Block & Bleed Feature		
16	Ball Support	Trunnion Mounted Ball		
17	Lifting Lugs	Required for valves weighing 250kg or heavier		
18	Sealant Injection Facility	Not required		
19	Vent / Drain Requirements	Not required		
20	Locking Facility	Required		
21	Stem	Anti-Blow-Out Type		
22	Seat	Self-Relieving		
23	Cavity Pressure Relief	Self-Relieving Seat		
24	Anti-Static Design	Required		
25	**Materials**			
26	Body	ASTM A105N		
27	Cover / Bonnet	ASTM A105N		
28	Ball (Solid)	A105N + ENP		
29	Stem / Trim	A105N + ENP		
30	Seat Housing	ASTM A182 F6A (13CR)		
31	Seat Insert	PEEK (Note 5.11)		
32	Primary / Secondary Seals	Elastomer O-rings, AED (Note 5.30) (Note 5.11)		
33	Gland Packing	Filled Graphite (Note 5.2) (Note 5.11)		
34	Anti-Static Device / Spring	INCONEL 718		
35 36	Gaskets	RING JOINT, OCTAGONAL, CL900, ASME B16.20, SOFT IRON		
37 38	Bolting	ASTM A193 Gr. B7; A194 Gr. 2H (Hot Dip Zinc Galvanized) ***(Note 5.22)***		
39	Hand Lever	Malleable Steel / Nodular Iron		
40	**Codes and Standards / Testing / Certification**			
41	Valve Design Code	As Per API 6D/DEP 31.36.00.30-Gen		
42	End to End Dimensions	As Per API 6D		
43	End Connection	As Per ASME B16.5		
44	Fire Safe	As Per API 6FA / BS EN ISO 10497		
45	Hydrostatic & Pneumatic Test	As Per API 6D		
46 47	Low Pressure Gas Seat Test	Type II As Per API 6D (***at 6 barg*** with Acceptance Criteria as defined in Annex H.3.4)		
48	Leakage Test	Leak Rate "A" As Per ISO 5208		
49 50	NDT (MPI / DP / Radiography / UT)	As Per Section 9 of Specification for Mainline Valves (Doc No.: EA-BFPCL-GPMC-GGPL-FEED-PPL-DSH-003)		
51 52	Material Certification	As Per Section DEP 31.36.00.30-Gen. section 15.0		
53 54 55	Painting / External Coating	As Per Coating / Painting Materials Specification - Linepipes / Pipelines / Field Joints, Structures, and Equipment (Doc. No.: EA-BFPCL-GPMC-GGPL-FEED-MCM-SPC-010)		
56 57	Marking	As Per Section 10 of Specification for Mainline Valves (Doc No.: EA-BFPCL-GPMC-GGPL-FEED-PPL-SDSH003)		
58 59	Other Applicable Project Spec.	Specification for Mainline Valves (Doc No.: EA-BFPCL-GPMC-GGPL-FEED-PPL-DSH-003)		

60	**Table 2.1 (As Per Valve List)**									
61 62	**S/N**	**Location**	**Pipe Class**	**Corrossion Allowance**	**Size**	**Rating**	**End Connection**	**Valve Tag**	**Line Number**	**Reference P&ID Number**
63	1	BBOU Manifold Station		3.0mm	2"	900#	RTJ FLANGED		BBOU-24"-P-012-X70-CL900	EA-BFPCL-GPMC-GGPL-FEED-PSE-DWG-003_02
64	2	BBOU Manifold Station		3.0mm	2"	900#	RTJ FLANGED		BBOU-24"-P-012-X70-CL900	EA-BFPCL-GPMC-GGPL-FEED-PSE-DWG-003_02
65	3	BBOU Manifold Station		3.0mm	2"	900#	RTJ FLANGED		BBOU-24"-P-012-X70-CL900	EA-BFPCL-GPMC-GGPL-FEED-PSE-DWG-003_02
66	4	BBOU Manifold Station		3.0mm	2"	900#	RTJ FLANGED		BBOU-24"-P-012-X70-CL900	EA-BFPCL-GPMC-GGPL-FEED-PSE-DWG-003_02
67	5	Elepa Manifold Station		6.0mm	2"	900#	RTJ FLANGED	MV1055	ELEP-24"-P-012-X70-CL900	EA-BFPCL-GPMC-GGPL-FEED-PSE-DWG-003_05
68	6	Elepa Manifold Station		6.0mm	2"	900#	RTJ FLANGED	MV1040	ELEP-36"-P-011-X70-CL900	EA-BFPCL-GPMC-GGPL-FEED-PSE-DWG-003_06
69	7	Elepa Manifold Station		6.0mm	2"	900#	RTJ FLANGED	MV1041	ELEP-36"-P-011-X70-CL900	EA-BFPCL-GPMC-GGPL-FEED-PSE-DWG-003_06
70	8	Elepa Manifold Station		6.0mm	2"	900#	RTJ FLANGED	MV1042	ELEP-36"-P-011-X70-CL900	EA-BFPCL-GPMC-GGPL-FEED-PSE-DWG-003_06
71	9	Akabel Pigging Station		6.0mm	2"	900#	RTJ FLANGED	MV1020	AKABP-36"-P-012-X70-CL900	EA-BFPCL-GPMC-GGPL-FEED-PSE-DWG-003_05
72	10	Akabel Pigging Station		6.0mm	2"	900#	RTJ FLANGED	MV1019	AKABP-36"-P-012-X70-CL900	EA-BFPCL-GPMC-GGPL-FEED-PSE-DWG-003_05
73	11	Akabel Pigging Station		6.0mm	2"	900#	RTJ FLANGED	MV1018	AKABP-36"-P-012-X70-CL900	EA-BFPCL-GPMC-GGPL-FEED-PSE-DWG-003_05
74	12	Akabel Pigging Station		6.0mm	2"	900#	RTJ FLANGED	MV1055	AKABP-36"-P-012-X70-CL900	EA-BFPCL-GPMC-GGPL-FEED-PSE-DWG-003_05

S/N	Datasheet 3: Datasheet for Ball Valve, Carbon Steel, Flanged RTJ, CL 900, 4" and Above, FB			
1	**Description**	**Specification Requirement**	**Vendor's Confirmation**	
2	**General Data**			
3	Piping Class	As Per Table 3.1 Below		
4	Valve Type	Ball Valve		
5	Valve Size Range	4" and above		
6	Valve ASME Class / Rating	900#		
7	Valve Port	Full Bore		
8	Valve Service	Hydrocarbon Gas / Open / Close Drain		
9	Valve End Connection	Flanged RTJ		
10	Corrosion Allowance	As Per Table 3.1 Below		
11	Design Temperature (Min / Max)	- 29 / 150 °C		
12	Design Pressure	Full rating as per ASME B16.34		
13	**Design**			
14	Valve Operation	Gear Operated Hand Wheel		
15	Body Construction	Split Body, Long Pattern with Double Block & Bleed Feature		
16	Ball Support	Trunnion Mounted Ball		
17	Lifting Lugs	Required for valves weighing 250kg or heavier		
18	Sealant Injection Facility	Required for Size 6" and Above		
19	Vent / Drain Requirements	As per API 6D		
20	Locking Facility	Required		
21	Stem	Anti-Blow-Out Type		
22	Seat	Self-Relieving		
23	Cavity Pressure Relief	DBB As Per API 6D		
24	Anti-Static Design	Required		
25	**Materials**			
26	Body	ASTM A105N		
27	Cover / Bonnet	ASTM A105N		
28	Ball (Solid)	A105N + ENP		
29	Stem / Trim	A105N + ENP		
30	Seat Housing	ASTM A182 F6A (13CR)		
31	Seat Insert	PEEK (Note 5.11)		
32	Primary / Secondary Seals	Elastomer O-rings, AED (Note 5.30) (Note 5.11)		
33	Gland Packing	Filled Graphite (Note 5.2) (Note 5.11)		
34	Anti-Static Device / Spring	INCONEL 718		
35 36	Gaskets	RING JOINT, OCTAGONAL, CL900, ASME B16.20, SOFT IRON		
37 38	Bolting	ASTM A193 Gr. B7; A194 Gr. 2H (Hot Dip Zinc Galvanized) ***(Note 5.22)***		
39	Hand Lever	Malleable Steel / Nodular Iron		
40	**Codes and Standards / Testing / Certification**			
41	Valve Design Code	As Per API 6D/DEP 31.36.00.30-Gen		
42	Face to Face Dimensions	As Per API 6D		
43	End Connection	As Per ASME B16.5		
44	Fire Safe	As Per API 6FA / BS EN ISO 10497		
45	Hydrostatic & Pneumatic Test	As Per API 6D		
46 47	Low Pressure Gas Seat Test	Type II As Per API 6D (***at 6 barg*** with Acceptance Criteria as defined in Annex H.3.4)		
48	Leakage Test	Leak Rate "A" As Per ISO 5208		
49 50	NDT (MPI / DP / Radiography / UT)	As Per Section 9 of Specification for Mainline Valves (Doc No.: EA-BFPCL-GPMC-GGPL-FEED-PPL-DSH-003)		
51 52	Material Certification	As Per Section DEP 31.36.00.30-Gen. section 15.0		
53 54 55	Painting / External Coating	As Per Coating / Painting Materials Specification - Linepipes / Pipelines / Field Joints, Structures, and Equipment (Doc. No.: EA-BFPCL-GPMC-GGPL-FEED-MCM-SPC-010)		
56 57	Marking	As Per Section 10 of Specification for Mainline Valves (Doc No.: EA-BFPCL-GPMC-GGPL-FEED-PPL-DSH-003)		
58 59	Other Applicable Project Spec.	Specification for Mainline Valves (Doc No.: EA-BFPCL-GPMC-GGPL-FEED-PPL-DSH-003)		

60	Table 3.1 (As Per Valve List)									
61 62	S/N	Location	Pipe Class	Corrossion Allowance	Size	Rating	End Connection	Valve Tag	Line Number	Reference P&ID Number
63	1	BBOU Manifold Station		3.0mm	24"	900#	RTJ FLANGED	MV1039	BBOU-24"-P-012-X70-CL900	EA-BFPCL-GPMC-GGPL-FEED-PSE-DWG-003_02
64	2	Elepa Manifold Station		6.0mm	24"	900#	RTJ FLANGED	MV1057	ELEP-24"-P-012-X70-CL900	EA-BFPCL-GPMC-GGPL-FEED-PSE-DWG-003_05
65	3	Elepa Manifold Station		6.0mm	36"	900#	RTJ FLANGED	MV1039	ELEP-36"-P-011-X70-CL900	EA-BFPCL-GPMC-GGPL-FEED-PSE-DWG-003_06
66	4	Elepa Manifold Station		6.0mm	36"	900#	RTJ FLANGED	MV1043	ELEP-36"-P-011-X70-CL900	EA-BFPCL-GPMC-GGPL-FEED-PSE-DWG-003_06
67	5	Elepa Manifold Station		6.0mm	36"	900#	RTJ FLANGED	MV1044	ELEP-36"-P-011-X70-CL900	EA-BFPCL-GPMC-GGPL-FEED-PSE-DWG-003_06
68	6	Elepa Manifold Station		6.0mm	36"	900#	RTJ FLANGED	MV1045	ELEP-36"-P-011-X70-CL900	EA-BFPCL-GPMC-GGPL-FEED-PSE-DWG-003_06
69	7	Akabel Pigging Station		6.0mm	36"	900#	RTJ FLANGED	MV1057	AKABP-36"-P-012-X70-CL900	EA-BFPCL-GPMC-GGPL-FEED-PSE-DWG-003_05
70	8	Akabel Pigging Station		6.0mm	36"	900#	RTJ FLANGED	MV1045	AKABP-36"-P-012-X70-CL900	EA-BFPCL-GPMC-GGPL-FEED-PSE-DWG-003_05
71	9	Akabel Pigging Station		6.0mm	36"	900#	RTJ FLANGED	MV1044	AKABP-36"-P-012-X70-CL900	EA-BFPCL-GPMC-GGPL-FEED-PSE-DWG-003_05
72	10	Akabel Pigging Station		6.0mm	36"	900#	RTJ FLANGED	MV1043	AKABP-36"-P-012-X70-CL900	EA-BFPCL-GPMC-GGPL-FEED-PSE-DWG-003_05

S/N	Datasheet 4: Datasheet for Ball Valve, Carbon Steel, SDV, Flanged RTJ, CL 900, 4" and Above, FB		
1	**Description**	**Specification Requirement**	**Vendor's Confirmation**
2	**General Data**		
3	Piping Class	As Per Table 4.1 Below	
4	Valve Type	Ball Valve	
5	Valve Size Range	4" and above	
6	Valve ASME Class / Rating	900#	
7	Valve Port	Full Bore	
8	Valve Service	Hydrocarbon Gas / Open / Close Drain	
9	Valve End Connection	Flanged RTJ	
10	Corrosion Allowance	As Per Table 4.1 Below	
11	Design Temperature (Min / Max)	- 29 / 150 °C	
12	Design Pressure	Full rating as per ASME B16.34	
13	**Design**		
14	Valve Operation	Gear Operated Hand Wheel	
15	Body Construction	Split Body, Long Pattern with Double Block & Bleed Feature	
16	Ball Support	Trunnion Mounted Ball	
17	Lifting Lugs	Required for valves weighing 250kg or heavier	
18	Sealant Injection Facility	Required for Size 6" and Above	
19	Vent / Drain Requirements	As per API 6D	
20	Locking Facility	Required	
21	Stem	Anti-Blow-Out Type	
22	Seat	Self-Relieving	
23	Cavity Pressure Relief	DIB2 As Per API 6D	
24	Anti-Static Design	Required	
25	**Materials**		
26	Body	ASTM A105N	
27	Cover / Bonnet	ASTM A105N	
28	Ball (Solid)	A105N + ENP	
29	Stem / Trim	A105N + ENP	
30	Seat Housing	ASTM A182 F6A (13CR)	
31	Seat Insert	ASTM A182 F6A (13CR)	
32	Primary / Secondary Seals	Elastomer O-rings, AED (Note 5.30) (Note 5.11)	
33	Gland Packing	Filled Graphite (Note 5.2) (Note 5.11)	
34	Anti-Static Device / Spring	INCONEL 718	
35, 36	Gaskets	RING JOINT, OCTAGONAL, CL900, ASME B16.20, SOFT IRON	
37, 38	Bolting	ASTM A193 Gr. B7; A194 Gr. 2H (Hot Dip Zinc Galvanized) ***(Note 5.22)***	
39	Hand Lever	Malleable Steel / Nodular Iron	
40	**Codes and Standards / Testing / Certification**		
41	Valve Design Code	As Per API 6D / DEP 31.36.00.30-Gen	
42	Face to Face Dimensions	As Per API 6D	
43	End Connection	As Per ASME B16.5	
44	Fire Safe	As Per API 6FA / BS EN ISO 10497	
45	Hydrostatic & Pneumatic Test	As Per API 6D	
46, 47	Low Pressure Gas Seat Test	Type II As Per API 6D (***at 6 barg*** with Acceptance Criteria as defined in Annex H.3.4)	
48	Leakage Test	Leak Rate "A" As Per ISO 5208	
49, 50	NDT (MPI / DP / Radiography / UT)	As Per Section 9 of Specification for Mainline Valves (Doc No.: EA-BFPCL-GPMC-GGPL-FEED-PPL-DSH-003)	
51, 52	Material Certification	As Per Section DEP 31.36.00.30-Gen. section 15.0	
53, 54, 55	Painting / External Coating	As Per Coating / Painting Materials Specification - Linepipes / Pipelines / Field Joints, Structures, and Equipment (Doc. No.: EA-BFPCL-GPMC-GGPL-FEED-MCM-SPC-010)	
56, 57	Marking	As Per Section 10 of Specification for Mainline Valves (Doc No.: EA-BFPCL-GPMC-GGPL-FEED-PPL-DSH-003)	
58, 59	Other Applicable Project Spec.	Specification for Mainline Valves (Doc No.: EA-BFPCL-GPMC-GGPL-FEED-PPL-DSH-003)	

60	Table 4.1 (As Per Valve List)									
61 62	S/N	Location	Pipe Class	Corrossion Allowance	Size	Rating	End Connection	Valve Tag	Line Number	Reference P&ID Number
63	1	BBOU Manifold Station		3.0mm	24"	900#	RTJ FLANGED	SDV1002	BBOU-24"-P-012-X70-CL900	EA-BFPCL-GPMC-GGPL-FEED-PSE-DWG-003_02
64	2	Elepa Manifold Station		6.0mm	24"	900#	RTJ FLANGED	SDV1003	ELEP-24"-P-012-X70-CL900	EA-BFPCL-GPMC-GGPL-FEED-PSE-DWG-003_05
65	3	Elepa Manifold Station		6.0mm	36"	900#	RTJ FLANGED	SDV1002	ELEP-36"-P-011-X70-CL900	EA-BFPCL-GPMC-GGPL-FEED-PSE-DWG-003_06
66	4	Akabel Pigging Station		6.0mm	36"	900#	RTJ FLANGED	SDV1003	AKABP-36"-P-012-X70-CL900	EA-BFPCL-GPMC-GGPL-FEED-PSE-DWG-003_05

S/N	Datasheet 5: Datasheet for Globe Valve, Carbon Steel, Flanged RTJ, CL 900, 2" to 4"			
1	**Description**	**Specification Requirement**	**Vendor's Confirmation**	**Vendor's Confirmation**
2	**General Data**			
3	Piping Class	As Per Table 5.1 Below		
4	Valve Type	Tee Type Globe Valve		
5	Valve Size Range	2" to 4"		
6	Valve ASME Class/Rating	*900#*		
7	Valve Port	Regular Bore		
8 9	Valve Service	*Hydrocarbon Gas*		
10	Valve End Connection	*Flanged RTJ*		
11	Corrosion Allowance	As Per Table 5.1 Below		
12	Design Temperature (Min/Max)	- 29 / *150* °C		
13	Design Pressure	Full rating as per ASME B16.34		
14	**Design**			
15	Valve Operation	Handwheel		
16 17	Body Construction	Bolted Bonnet, T-Pattern, Swivel Plug Disc, *Stem Protective Cover & Position Indicator*		
18	Valve Support	Required for valves weighing 250kg or heavier		
19	Lifting Lugs	Required for valves weighing 250kg or heavier		
20	Stem	Rising Stem, OS&Y		
21	Seat / Back Seat / Stem Packing	Renewable		
22	Position Indicator	Required		
23	**Materials**			
24	Body	ASTM A216 Gr. WCB		
25	Cover / Bonnet	ASTM A216 Gr. WCB		
26	Disc	Trim 5		
27	Stem	Trim 5		
28	Gland Packing	Filled Graphite		
29	Seat Ring	Trim 5		
30	Primary / Secondary Seals	Elastomer O-rings, AED (Note 5.12)		
31	Gland Flange	ASTM A105N		
32 33	Gaskets	SPWD; SS304 Winding Graphite Filled; SS304 Inner Ring; CS Outer Ring		
34 35	Bolting	ASTM A193 Gr. B7; A194 Gr. 2H (Hot Dip Zinc Galvanized)		
36	Handwheel	Malleable Steel / Nodular Iron		
37	**Codes and Standards / Testing / Certification**			
38	Valve Design Code	As Per API 6D / DEP 31.36.00.30-Gen		
39	End to End Dimensions	As Per API 6D		
40	End Connection	As Per ASME B16.5		
41	Fire Safe	As Per API 6FA / BS EN ISO 10497		
42	Hydrostatic & Pneumatic Test	As Per API 598		
43	Low Pressure Gas Seat Test	As Per API 598 at ***6*** barg with Acceptance Criteria in Table 5		
44	Leakage Rate	Leak Rate "B" As Per ISO 5208		
45 46	NDT (MPI / DP / Radiography / UT)	As Per Section 9 of Specification for Mainline Valves (Doc No.: EA-BFPCL-GPMC-GGPL-FEED-PPL-DSH-003)		
47 48	Material Certification	As Per Section DEP 31.36.00.30-Gen. section 15.0		
49 50 51	Painting / External Coating	SAs Per Coating / Painting Materials Specification - Linepipes / Pipelines / Field Joints, Structures, and Equipment (Doc. No.: EA-BFPCL-GPMC-GGPL-FEED-MCM-SPC-010)		
52 53	Marking	As Per Section 10 of Specification for Mainline Valves (Doc No.: EA-BFPCL-GPMC-GGPL-FEED-PPL-DSH-003)		
54 55	Other Applicable Project Spec.	Specification for Mainline Valves (Doc No.: EA-BFPCL-GPMC-GGPL-FEED-PPL-DSH-003)		

56	Table 5.1 (As Per Valve List)								
57 58	**Location**	**Pipe Class**	**Corrossion Allowance**	**Size**	**Rating**	**End Connection**	**Valve Tag**	**Line Number**	**Reference P&ID Number**
65	BBOU Manifold Platform		3.0mm	2"	900#	RTJ FLANGED	E4R	BBOU-24"-P-012-X70-CL900	EA-BFPCL-GPMC-GGPL-FEED-PSE-DWG-003_02
76	Elepa manifold Station		6.0mm	2"	900#	RTJ FLANGED	E4R	ELEP-24"-P-012-X70-CL900	EA-BFPCL-GPMC-GGPL-FEED-PSE-DWG-003_05
77	Elepa manifold Station		6.0mm	2"	900#	RTJ FLANGED	E4R	ELEP-36"-P-011-X70-CL900	EA-BFPCL-GPMC-GGPL-FEED-PSE-DWG-003_06
83	Akabel Pigging Station		6.0mm	2"	900#	RTJ FLANGED	E4R	AKABP-36"-P-012-X70-CL900	EA-BFPCL-GPMC-GGPL-FEED-PSE-DWG-003_05